Chapter 12

Electrodynamics and Relativity

Problem 12.1

Problem 12.2

Problem 12.3

Problem 12.4

As the outlaws escape in their getaway car, which goes $\frac{3}{4}c$, the police officer fires a bullet from the pursuit car, which only goes $\frac{1}{2}c$. The muzzle velocity of the bullet (relative to the gun) is $\frac{1}{3}c$. Does the bullet reach its target (a) according to Galileo, (b) according to Einstein?

Solution:

(a) Adding up the velocities with Galileo's velocity addition rule, the resulting velocity of the bullet is

$$\frac{1}{2}c + \frac{1}{3}c = \frac{5}{6}c > \frac{3}{4}c$$

so the bullet does reach its target.

(b) Using Einstein's velocity addition rule, the result is instead

$$v_{\text{bullet}} = \frac{v_{\text{car}} + v_{\text{muzzle}}}{1 + \frac{v_{\text{car}} v_{\text{muzzle}}}{c^2}} = \frac{\frac{5}{6}c}{1 + \frac{1}{6}} = \frac{5}{7}c < \frac{3}{4}c$$

so the bullet does not reach its target. □

Problem 12.5

Synchronized clocks are stationed at regular intervals, a million km apart, along a straight line. When the clock next to you reads 12 noon:

(a) What time do you see on the 90th clock down the line?

(b) What time do you observe on that clock?

Solution:

(a) The 90th clock down the line is a distance $d = 90 \times 10^6$ km $= 90 \times 10^9$ m. The time it takes for light to travel this distance is

$$t = \frac{d}{c} = \frac{90 \times 10^9 \text{ m}}{3 \times 10^8 \text{ m/s}} = 300 \text{ s}.$$

This means that what you see on the 90th clock is what it showed 300 seconds ago, so the clock reads 11:55.

(b) The time you **observe** on that clock is taking into account the time it takes for the information to travel. Therefore, what you observe is what the clock really is, 12:00. □

Problem 12.6

Every 2 years, more or less, The New York Times publishes an article in which some astronomer claims to have found an object traveling faster than the speed of light. Many of these reports result from a failure to distinguish what is seen from what is observed — that is, from a failure to account for light travel time. Here's an example: A star is traveling with speed v at an angle θ to the line of sight. What is its apparent speed across the sky? (Suppose the light signal from b reaches the earth at a time Δt after the signal from a, and the star has meanwhile advanced a distance Δs across the celestial sphere; by "apparent speed," I mean $\frac{\Delta s}{\Delta t}$.) What angle θ gives the maximum apparent speed? Show that the apparent speed can be much greater than c, even if v itself is less than c.

Problem 12.7

In a laboratory experiment, a muon is observed to travel 800 m before disintegrating. A graduate student looks up the lifetime of a muon (2×10^{-6}s) and concludes that its speed was

$$v = \frac{800 \text{ m}}{2 \times 10^{-6}\text{s}} = 4 \times 10^8 \text{ m/s}.$$

Faster than light! Identify the student's error, and find the actual speed of this muon.

Solution:

The student did not take into account the lifetime of the muon in the muons reference time. The muons lifetime in its own reference time is

$$\gamma t = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} t$$

If it travels for 800m then its speed is

$$\frac{800 \text{ m}}{\gamma \times 2 \times 10^{-6} \text{s}} = v$$

Simplifying the distance over time gives

$$\frac{800 \text{ m}}{2 \times 10^{-6} \text{s}} = \frac{4}{3} c$$

Solving for v

$$\frac{4}{3} c = \gamma v = \frac{v}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

$$\downarrow$$

$$\left(\frac{4}{3} c\right)^2 = \frac{v^2}{1 - \left(\frac{v}{c}\right)^2}$$

$$\downarrow$$

$$\left(\frac{4}{3} c\right)^2 \left(1 - \left(\frac{v}{c}\right)^2\right) = v^2$$

$$\downarrow$$

$$\left(\frac{4}{3} c\right)^2 = v^2 \left(1 + \left(\frac{4}{3} c\right)^2 \frac{1}{c^2}\right) = v^2 \left(1 + \frac{16}{9}\right)$$

$$\downarrow$$

$$v = \frac{4}{3} c \frac{1}{\sqrt{1 + \frac{16}{9}}} = \frac{4}{3} c \frac{1}{\sqrt{\frac{25}{9}}} = \frac{12}{15} c = \boxed{\frac{4}{5} c}$$

□

Problem 12.8

A rocket ship leaves earth at a speed of $\frac{3}{5} c$. When a clock on the rocket says 1 hour has elapsed, the rocket ship sends a light signal back to earth.

(a) According to earth clocks, when was the signal sent?

(b) According to earth clocks, how long after the rocket left did the signal arrive back on earth?

(c) According to the rocket observer, how long after the rocket left did the signal arrive back on earth?

Solution:

(a) The time it takes for one hour to pass on the rocket, $t_r = 1\text{h}$, is according to earth the time t_e

$$t_e = \gamma t_r = \frac{1}{\sqrt{1-\frac{3^2}{5^2}}} t_r = \sqrt{\frac{25}{16}} t_r = \frac{5}{4} t_r = \frac{5}{4}\text{h}$$

(b) The time the signal from the rocket was sent, according to earth, was $\frac{5}{4}$h. This means the rocket has travelled a distance $d = t_e/(\frac{3}{5}c) = 4500\text{s} \times \frac{3}{5}c = 8.1 \times 10^{10}$ m. This distance takes 2700s to reach earth, with a total time of $4500 + 2700$ s $= 7200$ s $= 2$h

(c) We can use the result in (b), where the time on earth was 2 hours. According to the rocket, earth's clock run slow, so the time according to the rocket is

$$\gamma \times 2\text{h} = \frac{5}{4} \times 2\text{h} = 2.5\text{h}$$

Another solution, if the result for (b) is missing. Imagine that the rocket is now stationary and earth is moving away at $\frac{3}{5}c$. The signal is sent after 1 hour, and in this time the earth has travelled $60\text{min} \times \frac{3}{5}c = 6.48 \times 10^9$ m. This distance now needs to be covered by the signal, which takes $6.48 \times 10^9\text{m} \times \frac{1}{c} = 36\text{min}$. Repeating the argument, in those 36 minutes, the earth has travelled another distance, so we get $36\text{min} \times \frac{3}{5}c$. We can model this as an infinite sum, with the factor 3/5:

$$t = 60\text{min} + 36\text{min} \times \sum_0^\infty \left(\frac{3}{5}\right)^n$$

which is a geometric series with solution

$$60\text{min} + 36\text{min}\,\frac{1}{1-\frac{3}{5}} = 60\text{min} + 90\text{min} = 150\text{min} = 2\text{h}30\text{min}$$

□

Problem 12.9

A Lincoln Continental is twice as long as a VW Beetle, when they are at rest. As the Continental overtakes the VW, going through a speed trap, a (stationary) policeman observes that they both have the same length. The VW is going at half the speed of light. How fast is the Lincoln going?

Solution:

The length of the cars while stationary is denoted L. With length contraction due to the velocity accounted for the length is denoted L'. With v_{vw} given and $L_{\text{lc}} = 2L_{\text{vw}}$, the solution is obtained by setting $L'_{\text{lc}} = L'_{\text{vw}}$ and solving for v_{lc}.

$$L_{\text{lc}} = 2L_{\text{vw}}$$

$$L'_{\text{vw}} = L_{\text{vw}}\sqrt{1 - \frac{v_{\text{vw}}^2}{c^2}} = L_{\text{vw}}\frac{\sqrt{3}}{2}$$

$$L'_{\text{vw}} = L'_{\text{lc}} = L_{\text{lc}}\sqrt{1 - \frac{v_{\text{lc}}^2}{c^2}} = L_{\text{vw}}\frac{\sqrt{3}}{2}$$

Using $L_{\text{lc}} = 2L_{\text{vw}}$ and solving for v_{lc}, the velocity for which the length of the Lincoln Continental equals the length of the VW Beetle is

$$v_{\text{lc}} = \frac{\sqrt{13}}{4}c$$

□

Problem 12.10

A sailboat is manufactured so that the mast leans at an angle $\overline{\theta}$ with respect to the deck. An observer standing on a dock sees the boat go by at speed v. What angle does this observer say the mast makes?

Solution:

The direction of the speed is Lorentz contracted by $\frac{1}{\gamma}$. The horizontal and vertical components of the mast is

$$\overline{x} = \overline{l}\cos\left(\overline{\theta}\right), \quad \overline{y} = \overline{l}\sin\left(\overline{\theta}\right)$$

Transforming these components with regard to the velocity gives

$$x = l\frac{\cos(\theta)}{\gamma}, \quad y = l\sin(\theta)$$

This gives

$$\frac{\overline{y}}{\overline{x}} = \frac{y}{x}$$

$$\downarrow$$

$$\boxed{\tan\left(\overline{\theta}\right) = \gamma\tan(\theta)}$$

□

Problem 12.11

Problem 12.12

Solve

$$\text{(i) } \overline{x} = \gamma(x - vt)$$
$$\text{(ii) } \overline{y} = y$$
$$\text{(iii) } \overline{z} = z$$
$$\text{(iv) } \overline{t} = \gamma\left(t - \frac{v}{c^2}x\right)$$

for x, y, z, t in terms of $\overline{x}, \overline{y}, \overline{z}, \overline{t}$, and check that you recover

$$\text{(i') } x = \gamma(\overline{x} + v\overline{t})$$
$$\text{(ii') } y = \overline{y}$$
$$\text{(iii') } z = \overline{z}$$
$$\text{(iv') } t = \gamma\left(\overline{t} + \frac{v}{c^2}\overline{x}\right)$$

Solution:

From (i) and (iv) we have

$$\text{(i) } \quad \overline{x} = \gamma(x - vt) \rightarrow x = \frac{\overline{x}}{\gamma} + vt$$
$$\text{(iv) } \overline{t} = \gamma\left(t - \frac{v}{c^2}x\right) \rightarrow t = \frac{\overline{t}}{\gamma} + \frac{v}{c^2}x$$

Inserting t from (iv) into the expression for x in (i) gives

$$x = \frac{\overline{x}}{\gamma} + v\left(\frac{\overline{t}}{\gamma} + \frac{v}{c^2}x\right) \rightarrow x\left(1 - \frac{v^2}{c^2}\right) = \frac{\overline{x}}{\gamma} + \frac{v\overline{t}}{\gamma}$$
$$\downarrow$$
$$x\frac{1}{\gamma^2} = \frac{1}{\gamma}(\overline{x} + v\overline{t}) \rightarrow \boxed{x = \gamma(\overline{x} + v\overline{t}) = \text{(i')}}$$

Putting this expression for x into the expression for t in (iv) gives

$$t = \frac{\bar{t}}{\gamma} + \frac{v}{c^2}(\gamma(\bar{x} + v\bar{t})) = \gamma\left(\frac{\bar{t}}{\gamma^2} + \frac{v}{c^2}\bar{x} + \frac{v^2}{c^2}\bar{t}\right) = \gamma\left(\left(1 - \frac{v^2}{c^2} + \frac{v^2}{c^2}\right)\bar{t} + \frac{v}{c^2}\bar{x}\right)$$

$$\downarrow$$

$$\boxed{t = \gamma\left(\bar{t} + \frac{v}{c^2}\bar{x}\right) = \text{(iv')}}$$

(ii) $\longleftrightarrow$ (ii') and (iii) $\longleftrightarrow$ (iii') follow directly from the equations.

□

Problem 12.13

Sophie Zabar, clairvoyante, cried out in pain at precisely the instant her twin brother, 500 km away, hit his thumb with a hammer. A skeptical scientist observed both events (brother's accident, Sophie's cry) from an airplane traveling at $\frac{12}{13}c$ to the right. Which event occurred first, according to the scientist? How much earlier was it, in seconds?

Solution:

Using the Lorentz transformations

$$\bar{x} = \gamma(x - vt)$$
$$\bar{t} = \gamma\left(t - \frac{v}{c^2}x\right)$$

and setting Sophie's position to $x = 500\text{km}$, and her brothers position to $x = 0\text{km}$. The time of the brothers event according to the scientist is

$$\bar{t}_1 = \frac{1}{\sqrt{1 - \frac{12^2}{13^2}}}\left(0 - \frac{12}{13c}0\right) = 0\text{s}$$

The time of Sophie's event is

$$\bar{t}_2 = \frac{1}{\sqrt{1 - \frac{12^2}{13^2}}}\left(0 - \frac{12}{13c}500\text{ km}\right) = \frac{1}{\sqrt{\frac{25}{169}}}\left(-\frac{12}{13c}500\text{km}\right) =$$
$$= -\frac{13}{5}\frac{12}{13c}500\text{km} = -\frac{1200000}{c}\text{ m} = -0.004\text{ s} = -4\text{ms}$$

The twin brothers event happened 4ms earlier according to the scientist. □

Problem 12.14

Problem 12.15

Problem 12.16

Problem 12.17

Check

$$-\overline{a}^0\overline{b}^0 + \overline{a}^1\overline{b}^1 + \overline{a}^2\overline{b}^2 + \overline{a}^3\overline{b}^3 = -a^0b^0 + a^1b^1 + a^2b^2 + a^3b^3$$

using

$$\begin{cases} \overline{a}^0 = \gamma(a^0 - \beta a^1) \\ \overline{a}^1 = \gamma(a^1 - \beta a^0) \\ \overline{a}^2 = a^2 \\ \overline{a}^3 = a^3 \end{cases}$$

Solution:

Computing the first term gives

$$-\overline{a}^0\overline{b}^0 = -\gamma^2(a^0 - \beta a^1)(b^0 - \beta b^1) = -\gamma^2(a^0b^0 - a^0b^1\beta - a^1b^0\beta + a^1b^1\beta^2)$$

and the second term gives

$$\overline{a}^1\overline{b}^1 = \gamma^2(a^1 - \beta a^0)(b^1 - \beta b^0) = \gamma^2(a^1b^1 - a^1b^0\beta - a^0b^1\beta + a^0b^0\beta^2)$$

Adding these two terms gives

$$-\overline{a}^0\overline{b}^0 + \overline{a}^1\overline{b}^1 = \gamma^2(-a^0b^0 + a^0b^1\beta + a^1b^0\beta - a^1b^1\beta^2 + a^1b^1 - a^1b^0\beta - a^0b^1\beta + a^0b^0\beta^2) =$$

$$= \gamma^2(a^0b^0(\beta^2 - 1) + a^1b^1(1 - \beta^2)) = \gamma^2\left(-a^0b^0\frac{1}{\gamma^2} + a^1b^1\frac{1}{\gamma^2}\right) = \boxed{-a^0b^0 + a^1b^1}$$

The terms

$$\overline{a}^2\overline{b}^2 = a^2b^2$$
$$\overline{a}^3\overline{b}^3 = a^3b^3$$

Follow directly since $\overline{a}^2 = a^2, \overline{a}^3 = a^3$. ✓

□

Problem 12.18

(a) Write out the matrix that describes a Galilean transformation.

(b) Write out the matrix describing a Lorentz transformation along the y-axis.

(c) Find the matrix describing a Lorentz transformation with velocity v along the x-axis followed by a Lorentz transformation with velocity $\overline{v}$ along the y-axis. Does it matter in what order the transformations are carried out?

Solution:

(a) Mapping the time variable $t \to ct$ and setting $\beta = \frac{v}{c}$ gives the Galilean transformation

$$\begin{pmatrix} c\bar{t} \\ \overline{x} \\ \overline{y} \\ \overline{z} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -\beta & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

(b)

$$\begin{pmatrix} c\bar{t} \\ \overline{x} \\ \overline{y} \\ \overline{z} \end{pmatrix} = \begin{pmatrix} \gamma & 0 & -\gamma\beta & 0 \\ 0 & 1 & 0 & 0 \\ -\gamma\beta & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

(c) First writing out the transformation with velocity v along the x-axis, setting $\beta = \frac{v}{c}$

$$\Lambda_x = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

And now with velocity $\overline{v}$ in the y-axis, with $\overline{\beta} = \frac{\overline{v}}{c}$, $\overline{\gamma} = \frac{1}{\sqrt{1-\overline{\beta}^2}}$

$$\Lambda_y = \begin{pmatrix} \overline{\gamma} & 0 & -\overline{\gamma}\overline{\beta} & 0 \\ 0 & 1 & 0 & 0 \\ -\overline{\gamma}\overline{\beta} & 0 & \overline{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Transforming first along the x-axis and then the y-axis gives the matrix

$$\Lambda_y\Lambda_x = \begin{pmatrix} \overline{\gamma} & 0 & -\overline{\gamma}\overline{\beta} & 0 \\ 0 & 1 & 0 & 0 \\ -\overline{\gamma}\overline{\beta} & 0 & \overline{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \overline{\gamma}\gamma & -\overline{\gamma}\gamma\beta & -\overline{\gamma}\overline{\beta} & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ -\overline{\gamma}\overline{\beta}\gamma & \overline{\gamma}\overline{\beta}\gamma\beta & \overline{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Doing it the other way around gives another matrix:

$$\Lambda_x\Lambda_y = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \overline{\gamma} & 0 & -\overline{\gamma}\overline{\beta} & 0 \\ 0 & 1 & 0 & 0 \\ -\overline{\gamma}\overline{\beta} & 0 & \overline{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \gamma\overline{\gamma} & -\gamma\beta & -\gamma\overline{\gamma}\overline{\beta} & 0 \\ -\gamma\beta\overline{\gamma} & \gamma & \gamma\beta\overline{\gamma}\overline{\beta} & 0 \\ -\overline{\gamma}\overline{\beta} & 0 & \overline{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and we see that $\Lambda_y\Lambda_x \neq \Lambda_x\Lambda_y$

□

Problem 12.19

The parallel between rotations and Lorentz transformations is even more striking if we introduce the **rapidity**:

$$\theta \equiv \tanh^{-1}\left(\frac{v}{c}\right)$$

(a) Express the Lorentz transformation matrix Λ (Eq. 12.24) in terms of θ, and compare it to the rotation matrix (Eq. 1.29).

In some respects, rapidity is a more natural way to describe motion than velocity. For one thing, it ranges from $-\infty$ to ∞, instead of $-c$ to c. More significantly, rapidities add, whereas velocities do not.

(b) Express the Einstein velocity addition law in terms of rapidity.

Solution:

(a) Rewriting γ and β in terms of the rapidity gives

$$\tanh^{-1}\beta = \theta \to \beta = \tanh(\theta)$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} \to \gamma = \frac{1}{\sqrt{1-\tanh^2(\theta)}} = \frac{1}{\operatorname{sech}(\theta)} \to \gamma = \cosh(\theta)$$

This makes the transformation matrix

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \to \begin{pmatrix} \cosh(\theta) & -\cosh(\theta)\tanh(\theta) & 0 & 0 \\ -\cosh(\theta)\tanh(\theta) & \cosh(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Since $\tanh(\theta) = \frac{\sinh(\theta)}{\cosh(\theta)}$, the matrix can be further simplified to

$$\Lambda = \begin{pmatrix} \cosh(\theta) & -\sinh(\theta) & 0 & 0 \\ -\sinh(\theta) & \cosh(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Comparing it to the rotation matrix (Eq. 1.29) which is

$$\begin{pmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(b) The Einstein velocity addition law is

$$v_{AC} = \frac{v_{AB} + v_{BC}}{1 + \frac{v_{AB} v_{BC}}{c^2}}$$

$$\theta_{AB} = \tanh^{-1}\left(\frac{v_{AB}}{c}\right) \rightarrow \tanh(\theta_{AB}) = \frac{v_{AB}}{c}$$
$$\theta_{BC} = \tanh^{-1}\left(\frac{v_{BC}}{c}\right) \rightarrow \tanh(\theta_{BC}) = \frac{v_{BC}}{c}$$
$$\theta_{AC} = \tanh^{-1}\left(\frac{v_{AC}}{c}\right) \rightarrow \tanh(\theta_{AC}) = \frac{v_{AC}}{c}$$

This gives the Einstein velocity addition law in terms of the rapidity

$$c\tanh(\theta_{AC}) = \frac{c(\tanh(\theta_{AB}) + \tanh(\theta_{BC}))}{1 + \tanh(\theta_{AB})\tanh(\theta_{BC})}$$
$$\downarrow$$
$$\tanh(\theta_{AC}) = \frac{\tanh(\theta_{AB}) + \tanh(\theta_{BC})}{1 + \tanh(\theta_{AB})\tanh(\theta_{BC})}$$

Using the identity

$$\tanh(x + y) = \frac{\tanh(x) + \tanh(y)}{1 + \tanh(x)\tanh(y)}$$

gives

$$\tanh(\theta_{AC}) = \tanh(\theta_{AB} + \theta_{BC}) \rightarrow \boxed{\theta_{AC} = \theta_{AB} + \theta_{BC}}$$

□

Problem 12.20

(a) Event A happens at point $(x_A = 5, y_A = 3, z_A = 0)$ and at time t_A given by $ct_A = 15$; event B occurs at (10, 8, 0) and $ct_B = 5$, both in system $\mathcal{S}$.

(i) What is the invariant interval between A and B?

(ii) Is there an inertial system in which they occur simultaneously? If so, find its velocity (magnitude and direction) relative to $\mathcal{S}$.

(iii) Is there an inertial system in which they occur at the same point? If so, find its velocity relative to $\mathcal{S}$.

(b) Repeat part (a) for $A = (2, 0, 0), ct = 1$; and $B = (5, 0, 0), ct = 3$.

Solution:

(a) The displacement 4-vector is

$$\Delta x^\mu = x_A^\mu - x_B^\mu$$

The scalar product of Δx^μ with itself gives the invariant interval between two events:

$$I \equiv (\Delta x)^\mu (\Delta x)_\mu = -\left(\Delta x^0\right)^2 + \left(\Delta x^1\right)^2 + \left(\Delta x^2\right)^2 + \left(\Delta x^3\right)^2$$

The two events A and B are

$$\begin{gathered} x_A^\mu = (15, 5, 3, 0) \\ x_B^\mu = (5, 10, 8, 0) \\ \downarrow \\ \Delta x^\mu = (15 - 5, 5 - 10, 3 - 8, 0) = (10, -5, -5, 0) \end{gathered}$$

This gives the invariant interval

$$\begin{aligned} I = (\Delta x)^\mu (\Delta x)_\mu = (10, -5, -5, 0) \cdot (-10, -5, -5, 0) = \\ = -(10)^2 + (-5)^2 + (-5)^2 = -50 \end{aligned}$$

Since $I < 0$, the invariant interval is timelike, so there exist an inertial system in which they occur at the same point, but not at the same time.

Using the Lorentz transformation, we want to find the velocity of the frame in which they occur at the same point. This is equal to setting the difference in displacement to zero after Lorentz transformation.

$$\Delta \overline{x} = \gamma(\Delta x - v_x \Delta t) = 0 \rightarrow v_x = \frac{\Delta x}{\Delta t} = \frac{-5}{\frac{10}{c}} = -\frac{c}{2}$$

$$\Delta \overline{y} = \gamma\left(\Delta y - v_y \Delta t\right) = 0 \rightarrow v_y = \frac{\Delta y}{\Delta t} = \frac{-5}{10c} = -\frac{c}{2}$$

So the speed of the frame is

$$v = -\frac{c}{2}\hat{\mathbf{x}} - \frac{c}{2}\hat{\mathbf{y}}$$

(b) The displacement 4-vector is now

$$x_A^\mu = (1, 2, 0, 0)$$
$$x_B^\mu = (3, 5, 0, 0)$$
$$\downarrow$$
$$\Delta x^\mu = (1 - 3, 2 - 5, 0, 0)$$

This gives the invariant interval

$$I = (\Delta x)^\mu (\Delta x)_\mu = (-2, -3, 0, 0) \cdot (2, -3, 0, 0) =$$
$$= -(2)^2 + (-3)^2 = 5$$

Since $I > 0$, the invariant interval is spacelike so there exists a frame in which they occur at the same time, but not at the same point.

To find the velocity of the frame in which they occur at the same time, we set the difference in time to zero after Lorentz transformation.

$$\Delta \bar{t} = \gamma \left(\Delta t - \frac{v}{c^2} \Delta x \right) = 0 \to v = \frac{\Delta t}{\Delta x} c^2$$
$$v = \frac{ct_A - ct_B}{x_A - x_B} c^2 = \frac{1 - 3}{2 - 5} c = \frac{-2}{-3} c = \frac{2}{3} c$$

□

Problem 12.21

The coordinates of event A are $(x_A, 0, 0), t_A$, and the coordinates of event B are $(x_B, 0, 0), t_B$. Assuming the displacement between them is spacelike, find the velocity of the system in which they are simultaneous.

Solution:

To find the velocity of the frame in which two events happen at the same time, we set the difference in time to zero after Lorentz transformation.

$$\Delta \bar{t} = \gamma \left(\Delta t - \frac{v}{c^2} \Delta x \right) = 0 \to v = \frac{\Delta t}{\Delta x} c^2$$

$$\boxed{v = \frac{t_A - t_B}{x_A - x_B} c^2}$$

□

Problem 12.22

(a) Draw a space-time diagram representing a game of catch (or a conversation) between two people at rest, 10 ft apart. How is it possible for them to communicate, given that their separation is spacelike?

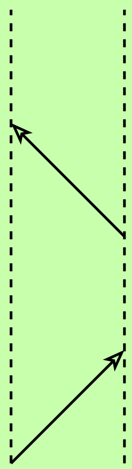
(b) There's an old limerick that runs as follows:

There once was a girl named Ms. Bright,
Who could travel much faster than light.
She departed one day,
The Einsteinian way,
And returned on the previous night.

What do you think? Even if she could travel faster than light, could she return before she set out? Could she arrive at some intermediate destination before she set out? Draw a space-time diagram representing this trip.

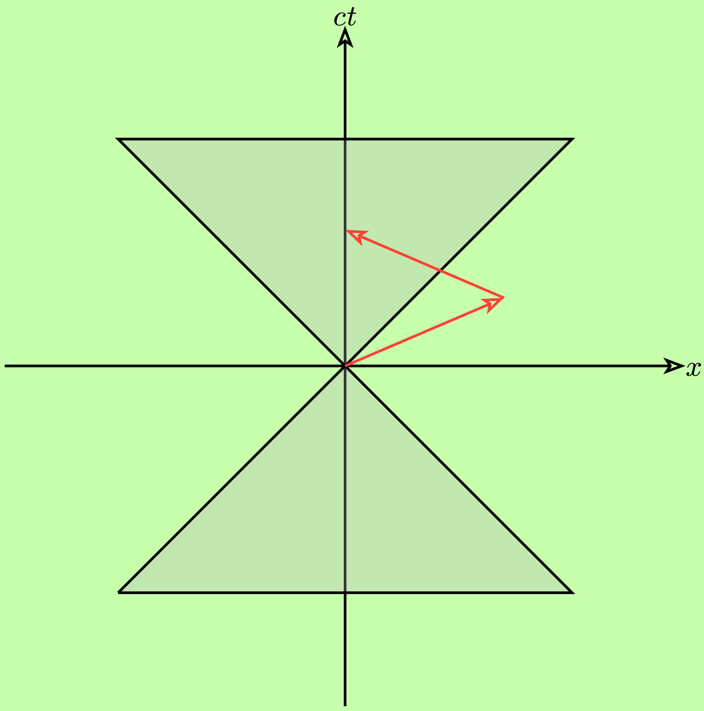
Solution:

(a)



You don't communicate with the person at the insant, but a future version of the person.

(b) No, travelling faster than light still makes you advance upwards in time.



(Unless it's possible to travel downwards?)

□

Problem 12.23

Inertial system $\overline{\mathcal{S}}$ moves in the x direction at speed $\frac{3}{5}c$ relative to system S. (The $\overline{x}$ axis slides along the x axis, and the origins coincide at $t = \overline{t} = 0$, as usual.)

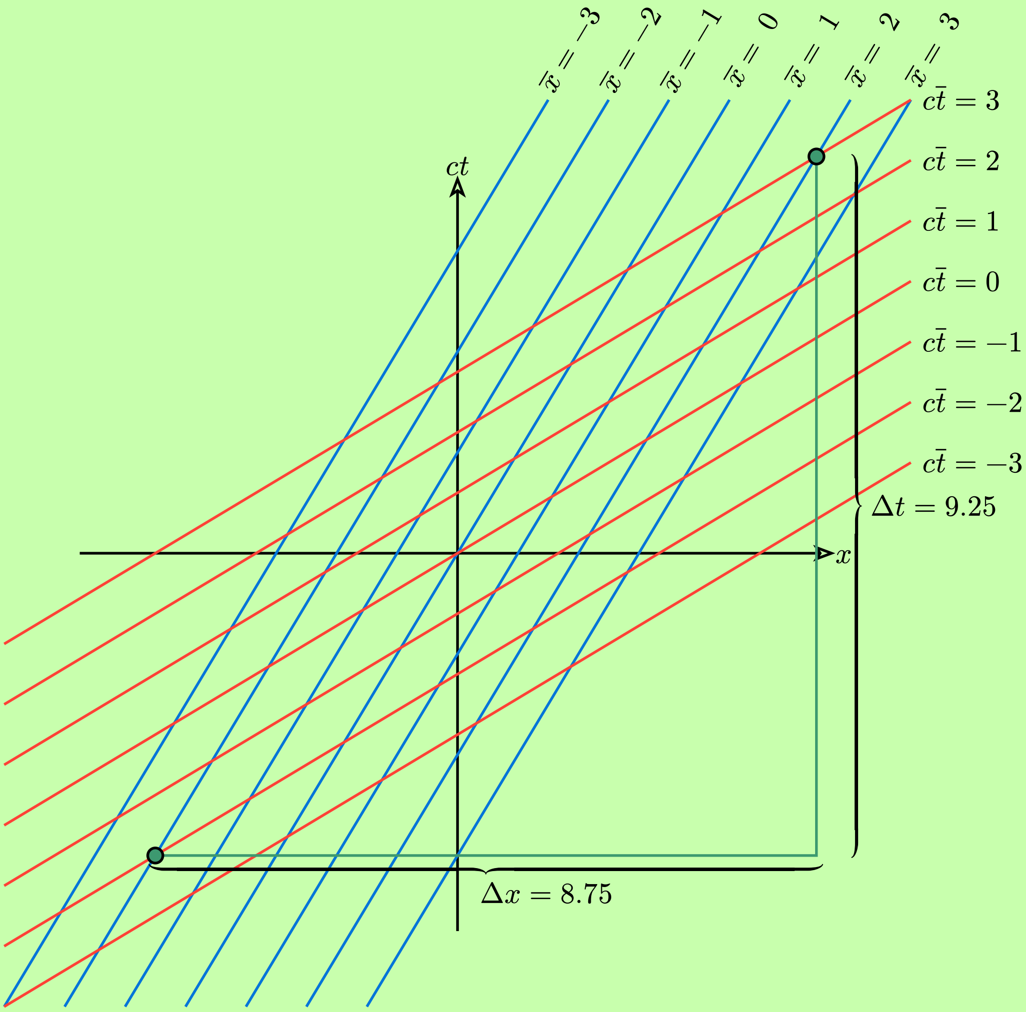
(a) On graph paper set up a Cartesian coordinate system with axes ct and x. Carefully draw in lines representing $\overline{x} = -3, -2, -1, 0, 1, 2,$ and 3. Also draw in the lines corresponding to $c\overline{t} = -3, -2, -1, 0, 1, 2,$ and 3. Label your lines clearly.

(b) In $\overline{\mathcal{S}}$, a free particle is observed to travel from the point $\overline{x} = -2$ at time $c\overline{t} = -2$ to the point $\overline{x} = 2$ at $c\overline{t} = 3$. Indicate this displacement on your graph. From the slope of this line, determine the particle's speed in $\mathcal{S}$.

(c) Use the velocity addition rule to determine the velocity in $\mathcal{S}$ algebraically.

Solution:

The graph will look like the following



(b) The displacement in $\mathcal{S}$ due to the particles movement in $\overline{\mathcal{S}}$, $\bar{x} : -2 \to 2, c\bar{t} : -2 \to 3$ is marked by the left $((\bar{x}, c\bar{t}) = (-2, -2))$ and right $((\bar{x}, c\bar{t}) = (2, 3))$ green dots. The slope of this line is $\frac{\Delta ct}{\Delta x} = \frac{9.25}{8.75} = \frac{37}{35}$, and the speed is given by the inverse of this slope, which is $\frac{35}{37}c$. Looking at the graph this makes immediate sense since the usual notion of speed is $\frac{\Delta x}{\Delta t}$ which in our case gives

$$\frac{8.75}{9.25}c = \boxed{\frac{35}{37}c}$$

(c) The speed of $\overline{\mathcal{S}}$ is $\frac{3}{5}c$, and from (b) we are given $\Delta c\bar{t} = 5, \Delta\bar{x} = 4 \to v_p = \frac{4}{5}c$ where v_p denotes the particles speed in the frame $\overline{\mathcal{S}}$. Einsteins velocity addition rule then gives

$$v = \frac{\frac{3}{5}c + \frac{4}{5}c}{1 + \frac{3}{5}c\frac{4}{5}c\frac{1}{c^2}} = \frac{\frac{7}{5}c}{\frac{25}{25} + \frac{12}{25}} = \frac{\frac{25\times 7}{5}}{25 + 12}c = \boxed{\frac{35}{37}c}$$

which is exactly the result in (b). □

Problem 12.24

(a) Equation 12.40 defines proper velocity in terms of ordinary velocity. Invert that equation to get the formula for u in terms of η.

(b) What is the relation between proper velocity and rapidity (Eq. 12.34)? Assume the velocity is along the x direction, and find η as a function of θ.

Solution:

(a) The proper velocity is

$$\boldsymbol{\eta} = \frac{1}{\sqrt{1-\frac{u^2}{c^2}}}\mathbf{u}$$

Squaring both sides gives

$$\eta^2 = \frac{1}{1-\frac{u^2}{c^2}}u^2$$

Solving for u

$$\eta^2\left(1-\frac{u^2}{c^2}\right) = u^2$$

$$\eta^2 = u^2 + \eta^2\frac{u^2}{c^2}$$

$$\eta^2 = u^2\left(1+\frac{\eta^2}{c^2}\right)$$

$$\frac{\eta^2}{1+\frac{\eta^2}{c^2}} = u^2$$

Now taking the square root on both sides gives the ordinary velocity u in terms of the proper velocity η

$$\boxed{\mathbf{u} = \frac{1}{\sqrt{1+\eta^2/c^2}}\boldsymbol{\eta}}$$

(b) The rapidity θ is defined as

$$\theta = \tanh^{-1}\left(\frac{u}{c}\right) \to \frac{u}{c} = \tanh(\theta), \quad u = c\tanh(\theta)$$

Rewriting the proper velocity in terms of rapidity then gives

$$\boldsymbol{\eta} = \frac{1}{\sqrt{1-\tanh^2(\theta)}}c\tanh(\theta)$$

Using the hyperbolic relations

$$(\text{i})\ 1-\tanh^2(\theta)=\text{sech}^2(\theta)=\frac{1}{\cosh^2(\theta)}$$

$$(\text{ii})\ \tanh(\theta)=\frac{\sinh(\theta)}{\cosh(\theta)}$$

we can rewrite the proper velocity as

$$\boldsymbol{\eta}=c\frac{\sinh(\theta)}{\cosh(\theta)\frac{1}{\cosh(\theta)}}=\boxed{c\sinh(\theta)}$$

$\square$

Problem 12.25

Problem 12.26

Find the invariant product of the 4-velocity with itself, $\eta^\mu\eta_\mu$. Is η^μ timelike, spacelike, or lightlike?

Solution:

The proper velocity is

$$\boldsymbol{\eta}=\frac{1}{\sqrt{1-\frac{u^2}{c^2}}}\mathbf{u}$$

This is the spatial component of a 4-vector which has the zeroth component

$$\eta^0=\frac{c}{\sqrt{1-\frac{u^2}{c^2}}}$$

The invariant product is

$$(\eta^0,\eta)\cdot(-\eta^0,\eta)=-\frac{c^2}{1-\frac{u^2}{c^2}}+\frac{u^2}{1-\frac{u^2}{c^2}}=-c^2\frac{1-\frac{u^2}{c^2}}{1-\frac{u^2}{c^2}}=\boxed{-c^2}$$

The invariant product is therefore timelike ($I<0$). $\square$

Problem 12.27

A cop pulls you over and asks what speed you were going. "Well, officer, I cannot tell a lie: the speedometer read 4×10^8 m/s." He gives you a ticket, because the speed limit on this highway is 2.5×10^8 m/s. In court, your lawyer (who, luckily, has studied physics) points out that a car's speedometer measures proper velocity, whereas the speed limit is ordinary velocity. Guilty, or innocent?

Solution:

The relation between proper velocity and ordinary velocity is

$$\boldsymbol{\eta} = \frac{1}{\sqrt{1-\frac{u^2}{c^2}}}\mathbf{u}$$
$$\downarrow$$
$$\mathbf{u} = \frac{1}{\sqrt{1+\frac{\eta^2}{c^2}}}\boldsymbol{\eta}$$

Solving for $\mathbf{u}$ with $\boldsymbol{\eta} = \frac{4}{3}c$ gives

$$\mathbf{u} = \frac{1}{\sqrt{1+\frac{4^2}{3^2}}}\frac{4}{3}c = \frac{1}{\sqrt{\frac{25}{9}}}\frac{4}{3}c = \frac{3}{5}\frac{4}{3}c = \frac{4}{5}c = \boxed{2.4 \times 10^8 \text{ m/s}}$$

Since $\mathbf{u} < 2.5 \times 10^8$ m/s, the driver is innocent. □

Problem 12.28

Problem 12.29

(a) Repeat Prob. 12.2(a) using the (incorrect) definition $\mathbf{p} = m\mathbf{u}$, but with the (correct) Einstein velocity addition rule. Notice that if momentum (so defined) is conserved in $\mathcal{S}$, it is not conserved in $\overline{\mathcal{S}}$. Assume all motion is along the x axis.

(b) Now do the same using the correct definition, $\mathbf{p} = m\boldsymbol{\eta}$. Notice that if momentum (so defined) is conserved in $\mathcal{S}$, it is automatically also conserved in $\overline{\mathcal{S}}$. [Hint: Use Eq. 12.43 to transform the proper velocity.] What must you assume about relativistic energy?

Problem 12.30

If a particle's kinetic energy is n times its rest energy, what is its speed?

Solution:

The kinetic energy is given by

$$E_{\text{kin}} = mc^2 \left(\frac{1}{\sqrt{1 - \frac{u^2}{c^2}}} - 1 \right)$$

and the rest energy is

$$E_{\text{rest}} = mc^2$$

If we want the kinetic energy to be n times its rest energy, we set

$$\frac{1}{\sqrt{1 - \frac{u^2}{c^2}}} - 1 = n$$

Solving for u gives

$$\begin{aligned}
\frac{1}{1 - \frac{u^2}{c^2}} &= (n+1)^2 \\
\frac{1}{(n+1)^2} &= 1 - \frac{u^2}{c^2} \\
\frac{u^2}{c^2} &= 1 - \frac{1}{(n+1)^2} \\
u^2 &= \left(1 - \frac{1}{(n+1)^2}\right) c^2 \\
u &= \sqrt{\frac{(n+1)^2 - 1}{(n+1)^2}} c \\
u &= \frac{\sqrt{n^2 + 2n}}{(n+1)^2} c \\
& \boxed{u = \frac{\sqrt{n(n+2)}}{n+1} c}
\end{aligned}$$

□

Problem 12.31

Problem 12.32

Find the velocity of the muon in Ex. 12.8.

Solution:

$$\begin{aligned} E_{\text{before}} &= m_\pi c^2, & \mathbf{p}_{\text{before}} &= 0 \\ E_{\text{after}} &= E_\mu + E_\nu, & \mathbf{p}_{\text{after}} &= \mathbf{p}_\mu + \mathbf{p}_\nu \end{aligned}$$

Conservation of energy gives

$$m_\pi c^2 = E_\mu + E_\nu$$

while conservation of momentum gives

$$\mathbf{p}_\mu = -\mathbf{p}_\nu$$

From example 12.8 we got

$$E_\mu = \frac{(m_\pi^2 + m_\mu^2)c^2}{2m_\pi}$$

and we also have

$$E_\mu = \frac{m_\mu c^2}{\sqrt{1 - \frac{u_\mu^2}{c^2}}}$$

With these two equations we can solve for u_μ

$$\frac{(m_\pi^2 + m_\mu^2)c^2}{2m_\pi} = \frac{m_\mu c^2}{\sqrt{1 - \frac{u_\mu^2}{c^2}}}$$

$$\downarrow$$

$$1 - \frac{u_\mu^2}{c^2} = \frac{4m_\pi^2 m_\mu^2}{\left(m_\pi^2 + m_\mu^2\right)^2}$$

$$\downarrow$$

$$u_\mu^2 = c^2 - \frac{4m_\pi^2 m_\mu^2 c^2}{\left(m_\pi^2 + m_\mu^2\right)^2} = \frac{\left(m_\pi^2 + m_\mu^2\right)^2 - \left(4m_\pi^2 m_\mu^2\right)}{\left(m_\pi^2 + m_\mu^2\right)^2} c^2 =$$

$$= \frac{m_\pi^4 + 2m_\pi^2 m_\mu^2 + m_\mu^4 - 4m_\pi^2 m_\mu^2}{\left(m_\pi^2 + m_\mu^2\right)^2} c^2 = \frac{m_\pi^4 + m_\mu^4 - 2m_\pi^2 m_\mu^2}{\left(m_\pi^2 + m_\mu^2\right)^2} c^2 =$$

$$= \frac{\left(m_\pi^2 - m_\mu^2\right)^2}{\left(m_\pi^2 + m_\mu^2\right)^2} c^2$$

$$\downarrow$$

$$\boxed{u_\mu = \frac{m_\pi^2 - m_\mu^2}{m_\pi^2 + m_\mu^2} c}$$

□

Problem 12.33

A particle of mass m whose total energy is twice its rest energy collides with an identical particle at rest. If they stick together, what is the mass of the resulting composite particle? What is its velocity?

Solution:

A particle of mass m has rest energy

$$E_{\text{rest}} = mc^2$$

and total energy

$$E^2 - p^2c^2 = m^2c^4 \to E^2 = m^2c^4 + p^2c^2$$

Twice gives

$$\left(2mc^2\right)^2 = m^2c^4 + p^2c^2$$

Solving for (initial) momentum p

$$p = \sqrt{3}mc$$

The initial energy is the energy of the moving particle plus the energy of the stationary particle:

$$2mc^2 + mc^2 = 3mc^2$$

Both energy and momentum are conserved so we can write

$$E^2 - p^2c^2 = \left(3mc^2\right)^2 - \left(\sqrt{3}mc\right)^2 c^2 = 9m^2c^4 - 3m^2c^4 = 6m^2c^4$$

This means the mass of the composite particle is $6m^2$ which gives each particle a mass of

$$m_p = \sqrt{6m^2} = \sqrt{6}m$$

Using

$$E = \frac{m_p c^2}{\sqrt{1 - \frac{u^2}{c^2}}}$$

$$\downarrow$$

$$3mc^2 = \frac{\sqrt{6}mc^2}{\sqrt{1 - \frac{u^2}{c^2}}}$$

$$\downarrow$$

$$9\left(1 - \frac{u^2}{c^2}\right) = 6$$

$$\downarrow$$

$$1 - \frac{u^2}{c^2} = \frac{6}{9}$$

$$\downarrow$$

$$u = \sqrt{1 - \frac{6}{9}}c = \sqrt{\frac{1}{3}}c$$

$$\boxed{u = \frac{c}{\sqrt{3}}}$$

□

Problem 12.34

Problem 12.35

Problem 12.36

Problem 12.37

Problem 12.38

Problem 12.39

Problem 12.40

Problem 12.41

Problem 12.42

Problem 12.43

Problem 12.44

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Problem 12.72

Problem 12.73

Problem 12.74

Problem 12.75

Problem 12.76